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U.S. Patent No. 8,330,235 Infringement Chart

Title: Method to reduce mol damage on NiSi

Inventors: Karthik Ramani, Paul R. Besser

Abstract: Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

Claim	Tower '235 Accused Products
[1pre] A semiconductor device comprising:	To the extent the preamble is limiting, devices manufactured by Tower that rely on its RF processes (including its 65nm RF process) (hereinafter, “Tower ’235 Accused Products”) comprise a semiconductor device. For example, the Tower ’235 Accused Products include the Qorvo QM81026 chip.¹ That is, on information and belief, the QM81026 has substantially similar structure, function, operation, and implementation with respect to the claims of the ’235 patent as any other chip manufactured by Tower using its 65nm processes.

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower ’235 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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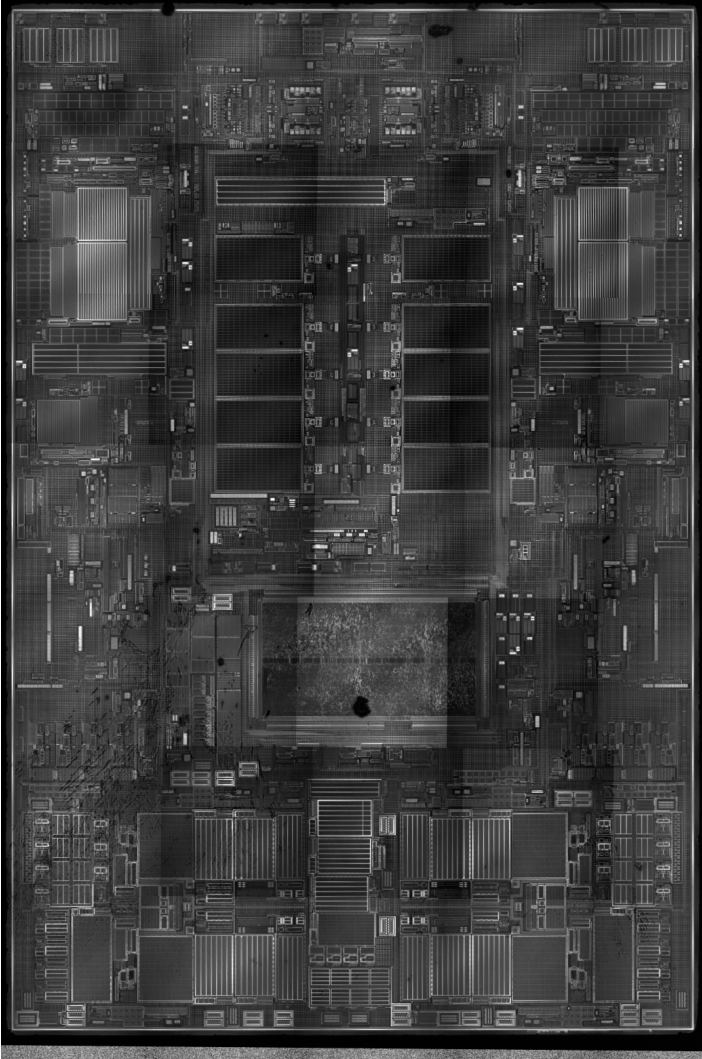
Claim	Tower '235 Accused Products
	
[1a] a silicon substrate,	The Tower '235 Accused Products comprise a silicon substrate.

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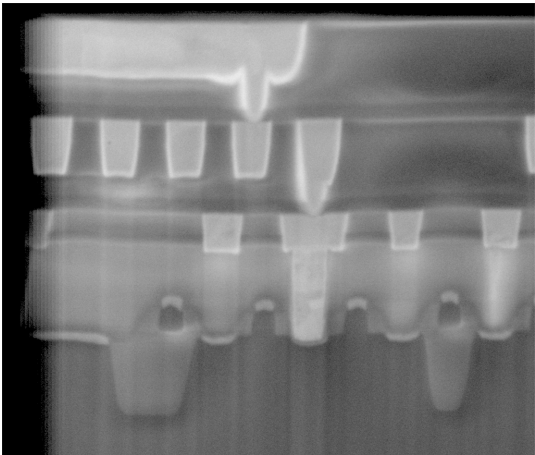
Claim	Tower '235 Accused Products
	For example, the reverse engineering analysis below shows a representative cross-section of the Tower '235 Accused Products, which shows a silicon substrate (bottom). 
[1b] a transistor, comprising source/drain regions and a gate electrode, on the substrate; and	The Tower '235 Accused Products comprise a transistor, comprising source/drain regions and a gate electrode, on the substrate. For example, the transistors in Tower '235 Accused Products each include source/drain regions and a gate electrode.

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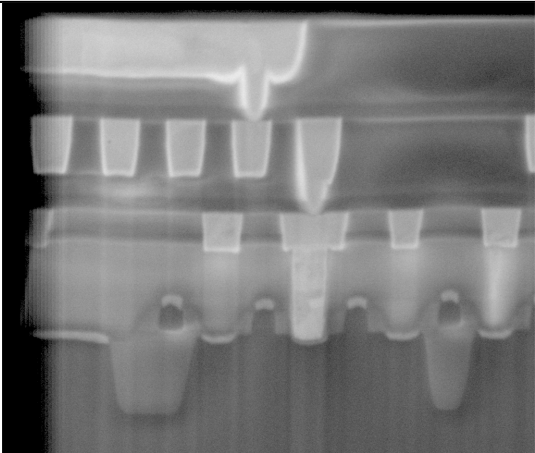
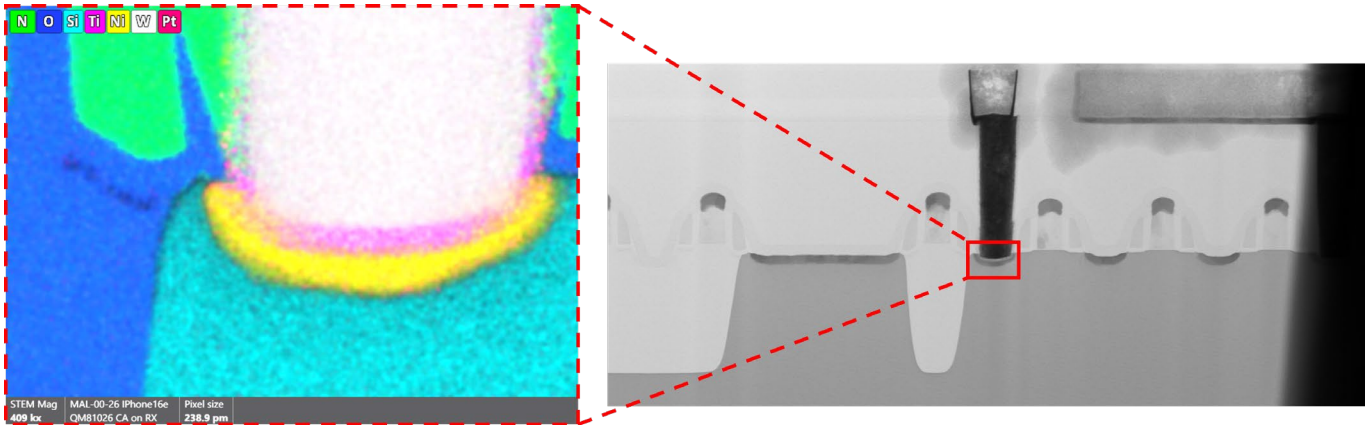
Claim	Tower '235 Accused Products
	
[1c] a nickel silicide layer on the gate electrode and source/drain regions,	The Tower '235 Accused Products comprise a nickel silicide layer on the gate electrode and source/drain regions. For example, the Tower '235 Accused Products include a nickel silicide layer on the gate electrode and source/drain regions. The reverse engineering analyses below show the presence of a nickel silicide layer on representative examples of the electrodes.  STEM Mag 400 kx MAL-00-26 iPhone1Ge QM81026 CA on RX Pixel size 230.9 nm

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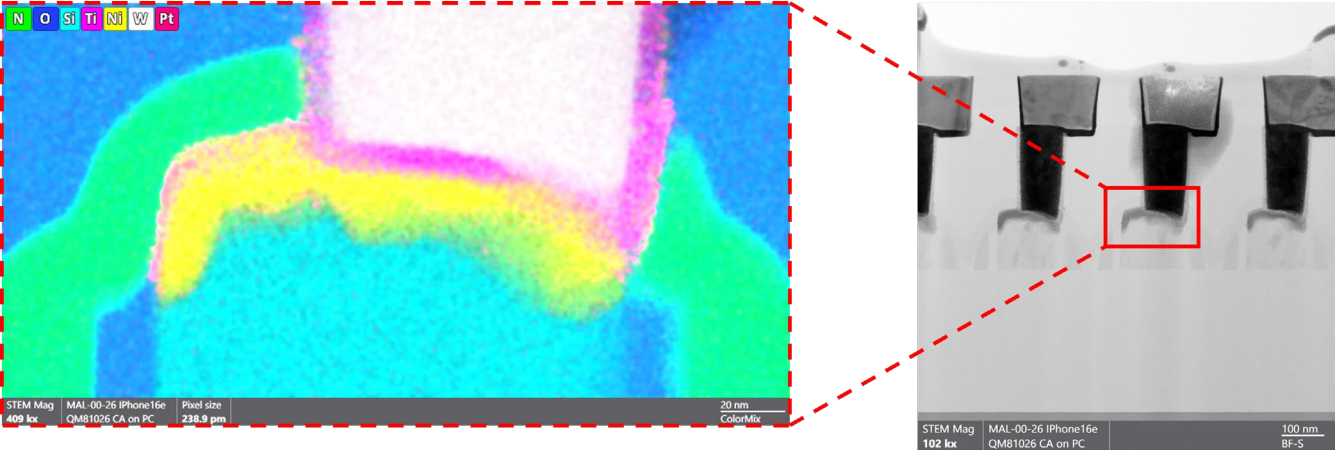
Claim	Tower '235 Accused Products
	 The figure consists of two side-by-side STEM images. The left image is a color map showing a composition gradient across a cross-section of a device. A legend at the top left identifies the elements: N (blue), O (green), Si (red), Ti (yellow), Ni (cyan), W (magenta), and Pt (black). The image shows a central region with a high concentration of Pt (black) and Ni (cyan), surrounded by other materials. The right image is a grayscale map of the same region, showing the same cross-section. A red box in the right image highlights a specific area, which is magnified in the left image. Both images have a scale bar at the bottom right indicating 20 nm. The left image also has a scale bar at the bottom left indicating 238.9 nm. The right image has a scale bar at the bottom right indicating 100 nm.
[1d] wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.	The Tower '235 Accused Products comprise the nickel silicide layer comprising platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface. For example, the Tower '235 Accused Products include a nickel silicide layer comprising platinum that increases in a direction away from the source/drain regions and the gate electrode (i.e., in an upward direction in the reverse engineering analyses below) from between about 8 wt. % and about 12 wt. % at the silicon interface. The reverse engineering analysis below shows the platinum composition gradient in the nickel silicide layer. The green arrow in the image below corresponds to the green arrow in the following two figures (i.e., such that the upward direction in the image below corresponds to the right-to-left direction in the following two materials analysis charts).

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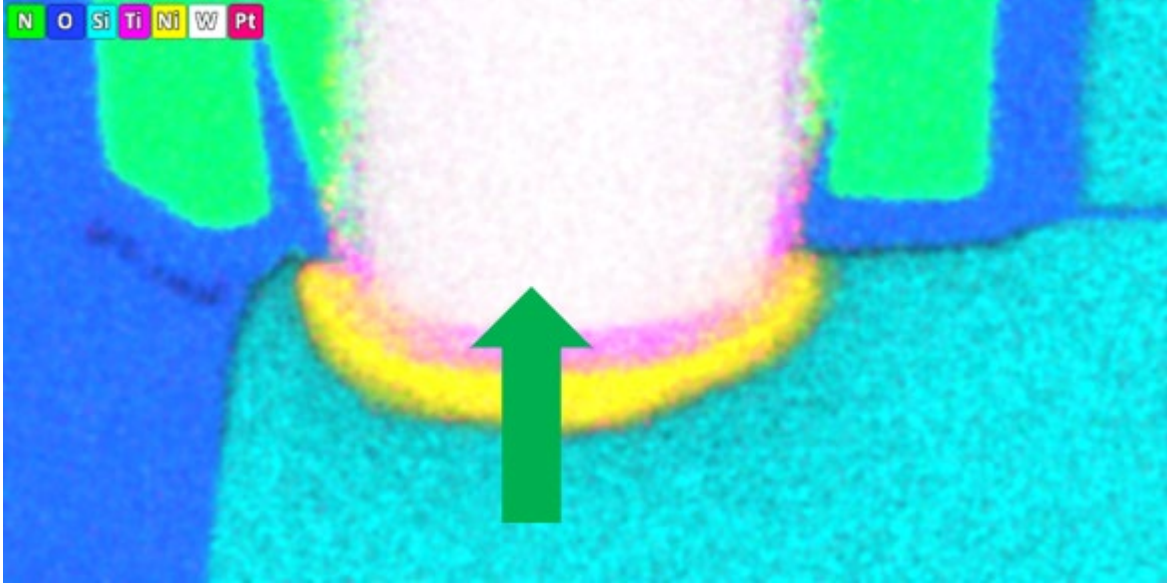
Claim	Tower '235 Accused Products
	An EDS line scan image showing elemental concentration profiles. The x-axis represents the scan line, and the y-axis represents the concentration of various elements. The elements listed at the top are N (Nitrogen), O (Oxygen), Si (Silicon), Ti (Titanium), Ni (Nickel), W (Tungsten), and Pt (Platinum). The image shows a cross-section of a material with a central region of high platinum concentration (yellow/orange) and a surrounding region of lower platinum concentration (blue/cyan). A green arrow points upwards, indicating an increase in platinum concentration from right to left. As shown below, the platinum concentration increases from between about 8 wt. % and 12 wt. % in an upward direction (i.e., right-to-left) in the image below.

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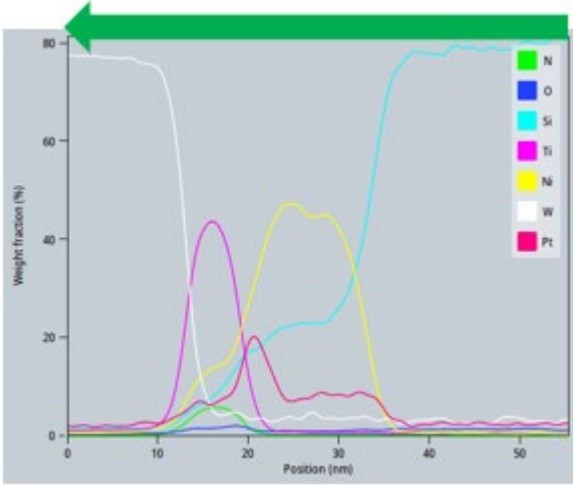
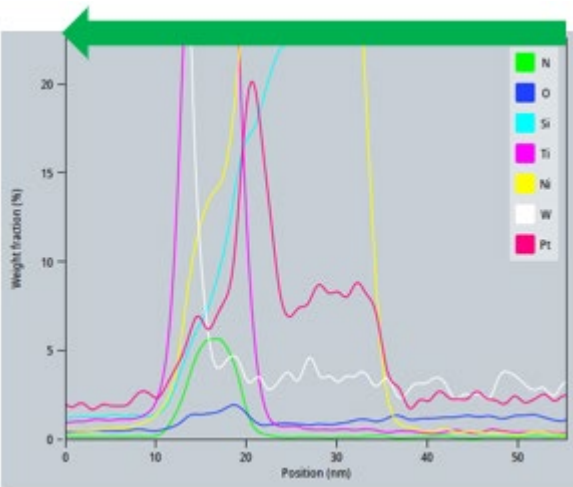
Claim	Tower '235 Accused Products
	 This chart displays the weight fraction (%) of various elements as a function of position (nm) from 0 to 50 nm. The y-axis ranges from 0 to 80%. A green arrow at the top points left, indicating the direction of the reverse engineering analysis. The legend includes N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (pink). The Si profile starts at ~75% at 0 nm and drops to 0 by ~15 nm. The Ti profile peaks at ~45% around 15 nm. The Ni profile peaks at ~45% around 25 nm. The Pt profile peaks at ~20% around 20 nm. The W profile is low and broad, peaking around 10 nm.  This chart displays the weight fraction (%) of various elements as a function of position (nm) from 0 to 50 nm. The y-axis ranges from 0 to 20%. A green arrow at the top points left, indicating the direction of the reverse engineering analysis. The legend includes N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (pink). The Si profile starts at ~20% at 0 nm and drops to 0 by ~15 nm. The Ti profile peaks at ~20% around 15 nm. The Ni profile peaks at ~20% around 25 nm. The Pt profile peaks at ~20% around 20 nm. The W profile is low and broad, peaking around 10 nm. Likewise, the reverse engineering analysis below show the platinum composition gradient in the nickel silicide layer. Again, the green arrow in the reverse engineering analysis below corresponds to the green arrow in the following two material analysis charts (i.e., such that the upward direction in the reverse

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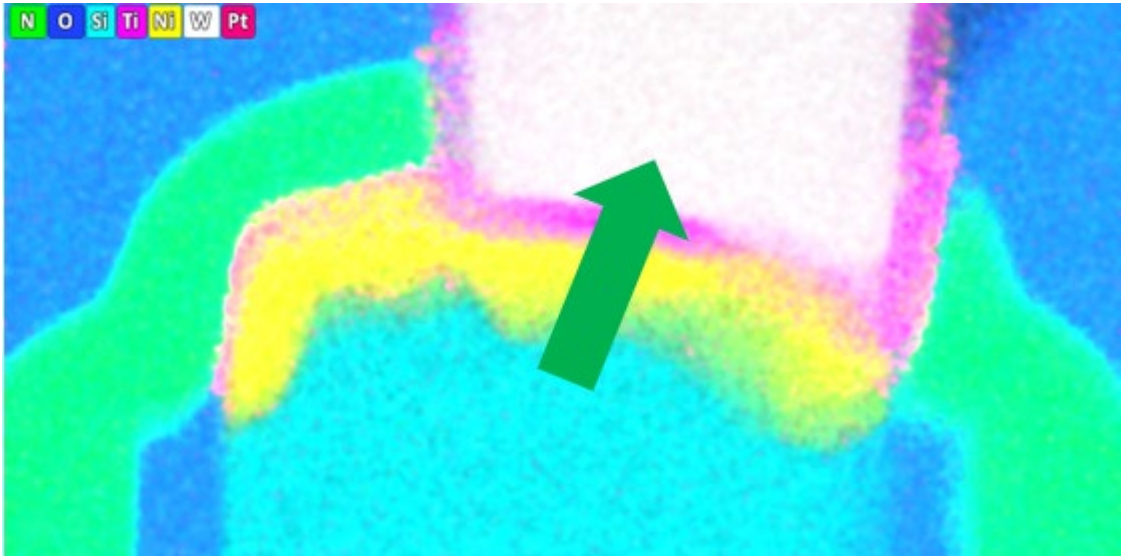
Claim	Tower '235 Accused Products
	engineering analysis immediately below corresponds to the right-to-left direction in the following two material analysis charts). A material analysis chart showing a cross-section of a material. The chart is color-coded to represent different elements: Nitrogen (N) in green, Oxygen (O) in blue, Silicon (Si) in cyan, Titanium (Ti) in magenta, Nickel (Ni) in yellow, Tungsten (W) in white, and Platinum (Pt) in red. A green arrow points from left to right across the center of the image, indicating the direction of engineering analysis. The chart shows a complex interface between different materials, with a prominent yellow region (Ni) and a magenta region (Ti) separated by a boundary. The platinum concentration increases from between about 8 wt. % and 12 wt. % at the silicon interface.

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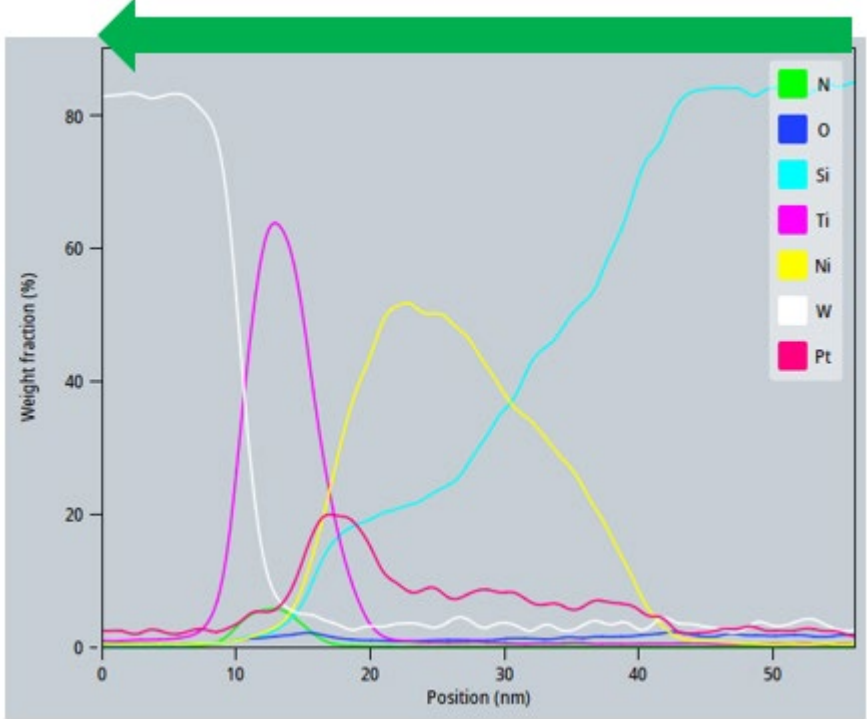
Claim	Tower '235 Accused Products																																																																																								
	 The chart displays the weight fraction (%) of several elements as a function of position (nm). The y-axis ranges from 0 to 80, and the x-axis ranges from 0 to 50 nm. The elements tracked are N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (pink). A large green arrow points from the right side of the chart towards the left, indicating a direction of analysis or a specific region of interest. <table><tr><th>Position (nm)</th><th>N (%)</th><th>O (%)</th><th>Si (%)</th><th>Ti (%)</th><th>Ni (%)</th><th>W (%)</th><th>Pt (%)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>85</td><td>0</td></tr><tr><td>10</td><td>0</td><td>0</td><td>0</td><td>10</td><td>0</td><td>80</td><td>0</td></tr><tr><td>15</td><td>0</td><td>0</td><td>0</td><td>65</td><td>0</td><td>10</td><td>0</td></tr><tr><td>20</td><td>0</td><td>0</td><td>0</td><td>10</td><td>40</td><td>5</td><td>20</td></tr><tr><td>25</td><td>0</td><td>0</td><td>0</td><td>0</td><td>50</td><td>5</td><td>10</td></tr><tr><td>30</td><td>0</td><td>0</td><td>0</td><td>0</td><td>40</td><td>5</td><td>10</td></tr><tr><td>35</td><td>0</td><td>0</td><td>10</td><td>0</td><td>20</td><td>5</td><td>10</td></tr><tr><td>40</td><td>0</td><td>0</td><td>60</td><td>0</td><td>5</td><td>5</td><td>5</td></tr><tr><td>45</td><td>0</td><td>0</td><td>85</td><td>0</td><td>0</td><td>5</td><td>0</td></tr><tr><td>50</td><td>0</td><td>0</td><td>85</td><td>0</td><td>0</td><td>5</td><td>0</td></tr></table> The material analysis chart immediately below illustrates a magnified graph (along the y-axis) of the material analysis chart immediately above.	Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)	0	0	0	0	0	0	85	0	10	0	0	0	10	0	80	0	15	0	0	0	65	0	10	0	20	0	0	0	10	40	5	20	25	0	0	0	0	50	5	10	30	0	0	0	0	40	5	10	35	0	0	10	0	20	5	10	40	0	0	60	0	5	5	5	45	0	0	85	0	0	5	0	50	0	0	85	0	0	5	0
Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)																																																																																		
0	0	0	0	0	0	85	0																																																																																		
10	0	0	0	10	0	80	0																																																																																		
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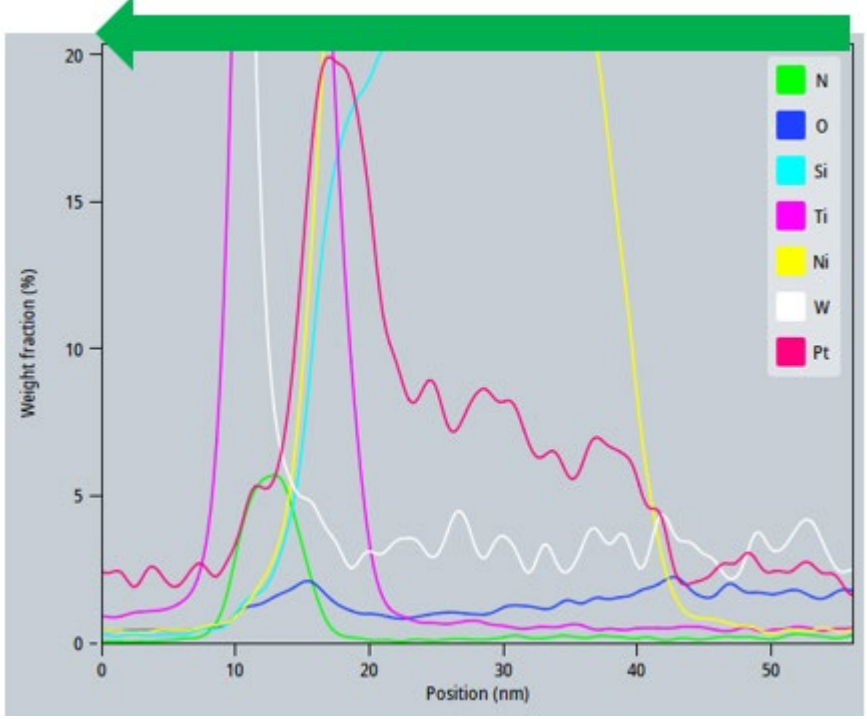
Claim	Tower '235 Accused Products
	 The graph displays the weight fraction (%) of various elements as a function of position (nm) from 0 to 55 nm. The elements tracked are Nitrogen (N, green), Oxygen (O, blue), Silicon (Si, cyan), Titanium (Ti, magenta), Nickel (Ni, yellow), Tungsten (W, white), and Platinum (Pt, red). A prominent peak for Silicon (Si) occurs at approximately 18 nm, reaching a weight fraction of about 20%. Other elements show smaller peaks or fluctuations across the range, with Platinum (Pt) showing a broad peak around 20-30 nm.
[10pre] A semiconductor device comprising:	The Tower '235 Accused Products comprise a semiconductor device. <i>See</i> limitation [1pre].
[10a] a silicon substrate;	The Tower '235 Accused Products comprise a silicon substrate. <i>See</i> limitation [1a].
[10b] a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode	The Tower '235 Accused Products comprise a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode, on the silicon substrate. For example, the Tower '235 Accused Products include both P-channel and N-channel transistors as part of the CMOS offering. Both the P-channel and N-channel transistors comprise source/drain regions and a gate electrode.

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Claim	Tower '235 Accused Products
electrode, on the silicon substrate; and	<i>See</i> limitation [1b].
[10c] a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and	The Tower '235 Accused Products comprise a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor. <i>See</i> limitation [1c].
[10d] a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,	The Tower '235 Accused Products comprise a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor. <i>See</i> limitation [1c].
[10e] wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from	In the Tower '235 Accused Products comprise, each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface. <i>See</i> limitation [1d]. The nickel silicide layers on the P-channel and N-channel transistors are substantially identical.

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between about 8 wt. % and about 12 wt. % at the silicon interface.	